

Pseudo-effectiveness of tangent bundles of smooth complete intersections and weighted hypersurfaces

(Joint work with Feng Shao)

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23 May 2024

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1. Background & Previous results (Basic Notation)

- We work over the field $\mathbb{C}$ of complex numbers.
- A smooth projective variety X is said to be **Fano**, if the anti-canonical divisor $-K_X$ is ample.
- The **(Fano) index** of a smooth Fano variety X is defined as the maximal integer i_X such that $-K_X \sim i_X H$ for some integral ample divisor H .
- It is known that $i_X \leq \dim(X) + 1$. If $i_X = \dim(X) + 1$, then $X \cong \mathbb{P}^n$ is a projective space; if $i_X = \dim(X)$, then $X \cong Q^n$ is a smooth quadric hypersurface in $\mathbb{P}^{n+1}$ (**Kobayashi-Ochiai, 1973**).
- A Fano manifold X is called **del Pezzo** if $i_X = \dim(X) - 1$. The **degree** of a del Pezzo manifold is defined as $H^{\dim(X)}$.
- Full classification ✓ for Fano manifolds of Fano index $\geq \dim(X) - 2$.
- The **tangent bundle** of a smooth projective variety is defined as the dual of its sheaf of Kähler differentials.

1. Background & Previous results (Ample & Strictly nef)

Theorem (Mori, 1979)=Hartshorne conjecture; cf. (Siu-Yau, 1980)

Every irreducible n -dimensional non-singular projective variety with ample tangent bundle defined over an algebraically closed field k of characteristic 0 is isomorphic to the projective space $\mathbb{P}_k^n$.

Definition (Positivity of vector bundles)

Let X be a smooth projective variety and E a vector bundle over X . Let $\mathbb{P}_X(E)$ be the Grothendieck projectivization. We say that E is ample (resp. nef, strictly nef, big, pseudo-effective, $\mathbb{Q}$ -effective) if so is $\mathcal{O}_{\mathbb{P}_X(E)}(1)$.

Theorem (Li-Ou-Yang, 2019)

A smooth complex projective variety with strictly nef tangent bundle is $\mathbb{P}^n$.

Both results ✓ for complex normal projective varieties (Sieder, 2017).

1. Background & Previous results (Nef)

Theorem (Demailly-Peternell-Schneider, 1994)

A compact Kähler manifold with nef tangent bundle, up to a finite étale cover, admits a smooth fibration over a complex torus whose fibre is a Fano manifold with nef tangent bundle.

Conjecture (Campana-Peternell, 1991)

Any Fano manifold whose tangent bundle is nef is rational homogeneous, or equivalently, it has globally generated tangent bundle.

- This conjecture is known up to dimension 5. (Campana-Peternell, Mok, Hwang, Watanabe, Kanemitsu).
- For projective manifolds with nef tangent bundles: every extremal contraction is smooth.
- A (weak) structure for projective klt varieties with (almost) nef tangent sheaf is known (Iwai-Matsumura-Z., 2023).

1. Background & Previous results (Big & Pseudo-effective)

A natural question

What about other positivity of tangent bundles, big or pseudo-effective tangent bundles?

Initial observations

- If T_X is big, then Miyaoka's generic semipositivity theorem $\Rightarrow X$ is covered by rational curves (Greb-Wong, 2020).
- If T_X is pseudo-effective, then X is not of general type; see e.g., (Höring-Peternell, 2021) or (Jia-Lee-Z., 2023).
- The property of tangent bundles being big/pseudo-effective descends along generically finite morphism between smooth projective varieties, but does not necessarily descend along fibrations.
- The quotient/determinant of a big/pseudo-effective vector bundle needs not be big/pseudo-effective, e.g., $\mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$.

1. Background & Previous results (Big & Pseudo-effective)

There are two well-developed approaches towards the classification on bigness/pseudo-effectiveness of tangent bundles:

(1). **Focus on X : twisted symmetric vector fields**; see (Druel, 2018) and (Höring-Liu-Shao, 2022). Let H be a big divisor on X .

- T_X is pseudo-effective $\Leftrightarrow \forall c > 0, \exists i, j \in \mathbb{N}$ with $i > cj$ such that $H^0(X, \text{Sym}^i T_X \otimes \mathcal{O}_X(jH)) \neq 0$.
- T_X is big $\Leftrightarrow \exists i, j \in \mathbb{N}$ such that $H^0(X, \text{Sym}^i T_X \otimes \mathcal{O}_X(-jH)) \neq 0$.

Examples

- (Hsiao, 2015) Toric varieties have big tangent bundles.
- (Höring-Liu-Shao, 2022) A smooth projective hypersurface of degree d has a big or pseudo-effective tangent bundle if and only if $d \leq 2$.
- (Kim, 2022) A projective bundle $\mathbb{P}_C(\mathcal{E})$ over a smooth curve C always has a pseudo-effective tangent bundle; it has a big tangent bundle if and only if $\mathcal{E}$ is not semi-stable or $C \cong \mathbb{P}^1$.

1. Background & Previous results (Big & Pseudo-effective)

(2). Focus on $\mathbb{P}(T_X)$: find certain special relative ample effective divisor; to show $\mathcal{O}_{\mathbb{P}(T_X)}(1)$ is big (resp. pseudo-effective), it is sufficient to show the following display

$$\mathcal{O}(1) \equiv a\pi^*H + bE$$

where $a, b > 0$ (resp. $a, b \geq 0$), E is effective and relative big, H is big (resp. pseudo-effective) on X and $\pi: \mathbb{P}(T_X) \rightarrow X$ is the projection.

Theorem (del Pezzo surfaces and del Pezzo threefolds); HLS22

- A Fano surface S of degree $d = K_S^2$ has big (resp. pseudo-effective) tangent bundle T_S if and only if $d \geq 5$ (resp. $d \geq 4$).
- Let X be a del Pezzo threefold such that $-K_X \sim 2H$ for some integral Cartier divisor H . Let $d = H^3$. Then
 - (1) T_X is big if and only if $d \geq 5$.
 - (2) Assume that X is general in its deformation family if $d = 2$. Then T_X is pseudo-effective if and only if $d \geq 4$.

1. Background & Previous results (Big & Pseudo-effective)

Theorem (Fano threefolds and higher dim del Pezzo manifolds)

- (Höring-Liu, 2023) Let X be a Fano threefold of Picard number 1. Then T_X is big if and only if X is (1) $\mathbb{P}^3$ or (2) Q^3 or (3) a quintic del Pezzo threefold V_5 .
- (Kim-kim-Lee, 2022) Let X be a Fano threefold of Picard number 2. Then T_X is big if and only if $-K_X^3 \geq 34$.
- (Beauville-Etesse-Höring-Liu-Voisin, 2023) A del Pezzo manifold of degree 4 has $\mathbb{Q}$ -effective but not big tangent bundle.
- (HLS22; Liu, 2023) A del Pezzo manifold of degree ≥ 5 has big tangent bundle.

Non-uniruled example (Höring-Peternell, 2021; Jia-Lee-Z., 2023)

A smooth non-uniruled projective surface with pseudo-effective tangent bundle is, up to a finite étale cover, either an abelian surface or a product of an elliptic curve and a high genus curve.

2. Main results & Proofs (Motivation)

Our goal in this project

Explore the tangent bundles of complete intersections/weighted hyper. so as to complete the classification of del Pezzo manifolds with big or pseudo-effective tangent bundles.

Let us recall the following classification (Fujita, Iskovskikh). Let (X, H) be an n -dim del Pezzo manifold of degree $d = H^n$. Assume that $n \geq 3$. Then

- $d = 1$ and X is a smooth weighted hypersurface of degree 6 in $\mathbb{P}(3, 2, 1^{\oplus n})$, alternatively, X admits a double cover over $\mathbb{P}(2, 1^{\oplus n})$.
- $d = 2$ and X is a smooth weighted hypersurface of degree 4 in $\mathbb{P}(2, 1^{\oplus(n+1)})$, alternatively, X admits a double cover over $\mathbb{P}^n$.
- $d = 3$ and X is a cubic hypersurface; T_X is not pseudo-effective.
- $d = 4$ and X is a complete intersection of two quadrics; T_X is $\mathbb{Q}$ -effective but not big.
- $d \geq 5$ and T_X is big.

2. Main results & Proofs (C.I. and Weighted-H.)

Theorem A (Liu, 2023; Shao-Z., 2023)

Let X be a non-linear smooth complete intersection of multi-degree $\underline{d} = (d_1, \dots, d_k)$ in a projective space. Then T_X is big $\Leftrightarrow \underline{d} = (2)$. Moreover, suppose that X is very general in its deformation family. Then T_X is pseudo-effective $\Leftrightarrow \underline{d} = (2)$ or $\underline{d} = (2, 2)$.

Theorem B (SZ24); cf. (HLS22)

Let X be a smooth weighted hypersurface of degree $d \geq a + b + 1$ in the weighted projective space $W = \mathbb{P}(a, b, 1^{\oplus n})(n \geq 3)$. Then

$$H^0(X, \text{Sym}^r(T_X \otimes \mathcal{O}_X(d - a - b - 1))) = 0, \quad \forall r \geq 1.$$

In particular, T_X is not big. Moreover, T_X is not pseudo-effective if $d \geq a + b + 2$.

It remains unclear when $d = a + b + 1$, i.e., T_X is not $\mathbb{Q}$ -effective $\Rightarrow?$ T_X is not pseudo-effective. ✓ for del Pezzo threefolds (HLS22)

2. Main results & Proofs (Del Pezzos of deg 1 or 2)

By Theorem B, one can remove the assumption of “being general in the deformation family” in (HLS22), i.e., a del Pezzo threefold of degree d has pseudo-effective tangent bundle if and only if $d \geq 4$.

Theorem C (SZ23 & 24)

Let X be a del Pezzo manifold of degree 1 or 2. Then the tangent bundle T_X is not big. Suppose further that X is very general in its deformation family. Then T_X is not pseudo-effective.

When X is very general, we obtain the corollary by combining (HLS22), (Liu, 23), (HL23), (Beauville-Etess-Höring-Liu-Voisin, 23).

Corollary (Classification of del Pezzo manifolds w/o certain positivity)

Let X be a del Pezzo manifold of degree d . Then T_X is big (resp. $\mathbb{Q}$ -effective) if and only if $d \geq 5$ (resp. $d \geq 4$). If X is further assumed to be very general in its deformation family in the case that $d = 1$ or 2 , then T_X is pseudo-effective if and only if $d \geq 4$.

2. Main results & Proofs (Finite covers over $\mathbb{P}^n$)

As a corollary of Theorem B, one can also obtain the following proposition.

Proposition (Finite covers over $\mathbb{P}^n$)

Let X be a smooth projective variety of dimension $n \geq 3$ and of Picard number 1. Suppose that X admits a finite cover of degree $s \geq 2$ over $\mathbb{P}^n$ which ramifies along a smooth hypersurface of degree d in $\mathbb{P}^n$.

- 1 If the pair $(s, d) \neq (2, 2)$, then the twisted tangent bundle $T_X(d - \frac{d}{s} - 2)H_X$ is not $\mathbb{Q}$ -effective.
- 2 T_X is big (resp. $\mathbb{Q}$ -effective) if and only if $(s, d) = (2, 2)$, i.e. X is isomorphic to a smooth hyperquadric.
- 3 T_X is not pseudo-effective if $(s, d) \neq (2, 2), (2, 4)$.

It follows from (Dimca, 1986) that $d = sr$ for some integer r and X is a hypersurface of degree sr in the weighted projective space $\mathbb{P}(r, 1^{\oplus(n+1)})$. Then we obtain that $H^0(X, \text{Sym}^m(T_X \otimes \mathcal{O}_X(sr - r - 2))) = 0$ for $m \geq 1$.

2. Main results & Proofs (Ingredients)

- Ingredient I (Höring-Peternell, 2024): Let $\mathcal{X} \rightarrow \Delta$ be a flat family such that each fibre is reduced and irreducible. If there exists one fibre whose tangent sheaf is not pseudo-effective, then so is the tangent sheaf of a very general fibre. In the smooth family case, this is known as the upper-semi continuity.
- Ingredient II (Liu, 23): Let $\mathcal{X} \rightarrow \Delta$ be a smooth family of Fano manifolds of Picard number 1 such that every fibre satisfies certain assumption. Suppose that $T_{\mathcal{X}_0}$ is big for one fibre. Then $T_{\mathcal{X}_t}$ is big for any $t \in \Delta$.

Therefore, to show Theorem A, we only need to find a special member, whose tangent bundle is not big (resp. not pseudo-effective).

Easy observation

Let $f : X \rightarrow Y$ be a finite surjective morphism between smooth projective varieties. Then we have the injection $0 \rightarrow T_X \rightarrow f^* T_Y$ and hence “ T_X being big/pseudo-effective $\Rightarrow T_Y$ being big/pseudo-effective”.

2. Main results & Proofs – Theorem A

Theorem A (Liu, 2023; Shao-Z., 2023)

Let X be a non-linear smooth complete intersection of multi-degree $\underline{d} = (d_1, \dots, d_k)$ in a projective space. Then T_X is big $\Leftrightarrow \underline{d} = (2)$. Moreover, suppose that X is very general in its deformation family. Then T_X is pseudo-effective $\Leftrightarrow \underline{d} = (2)$ or $\underline{d} = (2, 2)$.

- Let $X \subseteq \mathbb{P}^{n+k}$ be a smooth c.i. of $\dim n \geq 2$ and of multi-degree $(d_1, \dots, d_k)$ with $d_i \geq 2$ for any $1 \leq i \leq k$.
- By (HLS22), we may assume $k \geq 2$ and $n \geq 3$.
- Consider the following projection map:

$$\begin{aligned} \mathbb{P}^{n+k} \setminus \{p := [0 : \dots : 0 : 1]\} &\xrightarrow{\pi} \mathbb{P}^{n+k-1} \\ [x_0 : \dots : x_{n+k-1} : x_{n+k}] &\mapsto [x_0 : \dots : x_{n+k-1}]. \end{aligned}$$

Let $Y \subseteq \mathbb{P}^{n+k-1}$ be a general smooth complete intersection of multi-degree $(d_1, \dots, d_{k-1})$. Take $Y' \subseteq \mathbb{P}^{n+k}$ to be the closure of $\pi^{-1}(Y)$; then the ideal sheaf $\mathcal{I}_{Y'}$ of Y' is generated by the same polynomials as that of Y .

2. Main results & Proofs – Theorem A

Theorem A (Liu, 2023; Shao-Z., 2023)

Let X be a non-linear smooth complete intersection of multi-degree $\underline{d} = (d_1, \dots, d_k)$ in a projective space. Then T_X is big $\Leftrightarrow \underline{d} = (2)$. Moreover, suppose that X is very general in its deformation family. Then T_X is pseudo-effective $\Leftrightarrow \underline{d} = (2)$ or $\underline{d} = (2, 2)$.

- By Bertini's theorem, we take a degree d_k hypersurface H_k sufficiently general in $\mathbb{P}^{n+k}$ s.t. the scheme-theoretic intersection $X' := Y' \cap H_k$ is a smooth c.i. of multi-degree $(d_1, \dots, d_k)$.
- By construction, $X' \rightarrow Y$ is a finite cover of degree d_k and we reduce to $\underline{d} = (d_1)$ or $\underline{d} = (2, 2)$.
- The non-bigness then follows from (HLS22).
- The non-pseudo-effectiveness follows from (HLS22) and (Liu, 2023).

2. Main results & Proofs – Theorem B

Theorem B (SZ24); cf. (HLS22)

Let X be a smooth weighted hypersurface of degree $d \geq a + b + 1$ in the weighted projective space $W = \mathbb{P}(a, b, 1^{\oplus n})(n \geq 3)$. Then

$$H^0(X, \operatorname{Sym}^r(T_X \otimes \mathcal{O}_X(d - a - b - 1))) = 0, \quad \forall r \geq 1.$$

In particular, T_X is not big. Moreover, T_X is not pseudo-effective if $d \geq a + b + 2$.

To show the vanishing of twisted symmetric vector fields, we shall use:

$$\begin{aligned} \operatorname{Sym}^r(T_X \otimes \mathcal{O}_X(d - a - b - 1)) &= \operatorname{Sym}^r\left(\bigwedge^{n-1} \Omega_X \otimes \mathcal{O}_X(n-1)\right) \\ &= \mathbb{S}_\mu \Omega_X \otimes \mathcal{O}_X(r(n-1)), \end{aligned}$$

where $\mathbb{S}_\mu$ is the Schur functor corresponding to the partition $\mu = (r, \dots, r)$ with $(n-1)$ parts. By computing the Čech cohomology, we prove a result extending (Brückmann-Rackwitz, 1990).

2. Main results & Proofs – Theorem C (deg 2 case)

- Recall that X is a double cover of $\mathbb{P}^n$ branched along a smooth hypersurface of degree 4.
- It satisfies the additional assumption in (Liu, 2023). So to show the non-bigness, we only need to show a special member has non-big tangent bundle.
- Consider the del Pezzo manifold X of degree 2 defined by the following equation in the weighted projective space $\mathbb{P}(2, 1, \dots, 1)$

$$x_0^2 + x_1^4 + \dots + x_{n+1}^4 = 0.$$

- There is a finite morphism from X to the hyperquadric Q^n defined by the following

$$\begin{aligned} X &\longrightarrow Q^n = (x_0^2 + x_1^2 + \dots + x_{n+1}^2 = 0) \subset \mathbb{P}^{n+1} \\ (t_0 : t_1 : \dots : t_{n+1}) &\mapsto (t_0 : t_1^2 : \dots : t_{n+1}^2). \end{aligned}$$

Hence T_X is not big.

2. Main results & Proofs – Theorem C (deg 2 case)

- To prove the non-pseudo-effectiveness, we shall use the upper Semi-continuity and (Höring-Peternell, 2024).
- Let Y be a del Pezzo manifold of degree 3, i.e., a cubic hypersurface.

$$\begin{array}{ccccc} Y' & \longrightarrow & X' & \longrightarrow & W \\ \downarrow & & & & \\ Y & & & & \end{array}$$

Here, $Y' \rightarrow Y$ is the blow-up along a general point. It is easy to show that $-K_{Y'}$ is nef and big and it defines a generically finite morphism $Y' \rightarrow W = \mathbb{P}^{\dim(Y)}$. Let $Y' \rightarrow X'$ be the Stein factorization.

- By some classification result, X' is a singular del Pezzo variety of degree 2 and $Y' \rightarrow X'$ is isomorphic in codimension one.
- “ T_Y is not pseudo-effective” $\Rightarrow$ “ $T_{Y'}$ is not pseudo-effective” $\Leftrightarrow$ “ $T_{X'}$ is not pseudo-effective”.

3. Further discussion & Questions

Further Remark

- The same strategy with some minor modifications can be applied to del Pezzo degree 1 case.
- We expect those kinds of arguments work for **Mukai manifolds**, i.e., a smooth Fano variety X of dimension n whose anti-canonical divisor $-K_X$ is linearly equivalent to $(n-2)H_X$ for some integral ample divisor H_X ; the **genus** $g(X)$ is defined as $\frac{1}{2}H_X^n + 1$. It is known that if X is of Picard number 1, then $2 \leq g(X) \leq 12$ and $g(X) \neq 11$ hold.
- The troublesome case is when $g(X) = 12$, as the special member has non-trivial $\text{Aut}^0(X)$.

Proposition (SZ23)

Let X be a Mukai manifold of Picard number 1. If $g(X) \leq 5$, or $g(X) = 6$ and $\dim(X) \neq 6$, then the tangent bundle T_X is not big. If $g(X) \leq 5$ and X is assumed to be very general, then T_X is not pseudo-effective.

3. Further discussion & Questions

Question

Can we remove the very general assumption in Theorem C?

An observation due to (HLS22) and (Fu-Liu, 2022)

Let X be a Fano manifold of Picard number 1 and E a vector bundle over X . Then $W := \mathbb{P}_X(E)$ has Picard number 2. Assume that $\xi := \mathcal{O}_W(1)$ is pseudo-effective but not big.

- Its divisorial Zariski decomposition can be written as $\xi \equiv P + N$, where P is fixed and N is numerically movable in the sense that $N|_D$ is pseudo-effective for any prime divisor D on W .
- ξ is extremal $\Rightarrow$ either ξ is numerically movable or $\mathbb{Q}$ -effective.
- By Theorem B and taking E to be the tangent bundle of a del Pezzo manifold of degree 1 or 2, ξ is not $\mathbb{Q}$ -effective.
- To prove the non-pseudo-effectiveness, we only need to prove that it is not numerically movable.

Thank you for your attention!